

UI5con

How we replaced our legacy build infrastructure with UI5 Tooling

Florian Vogt & Merlin Beutlberger, SAP SE

July 6th, 2023

Build infrastructure for UI5

Internal production of the SAPUI5 product
(this includes OpenUI5)

Build infrastructure for UI5

Internal production of the SAPUI5 product (this includes OpenUI5)

70 libraries maintained
by over 400 developers

Produce artifacts for release via
ABAP, Java, CDN, npm, etc.

Build infrastructure for UI5

Internal production of the SAPUI5 product (this includes OpenUI5)

70 libraries maintained
by over 400 developers

Produce artifacts for release via
ABAP, Java, CDN, npm, etc.

Create source maps

Bundle resources

Minify resources

Compile themes

...

Generate metadata for
API reference, MTA build, npm
consumption, etc.

Build tools for UI5

Java UI5 Build

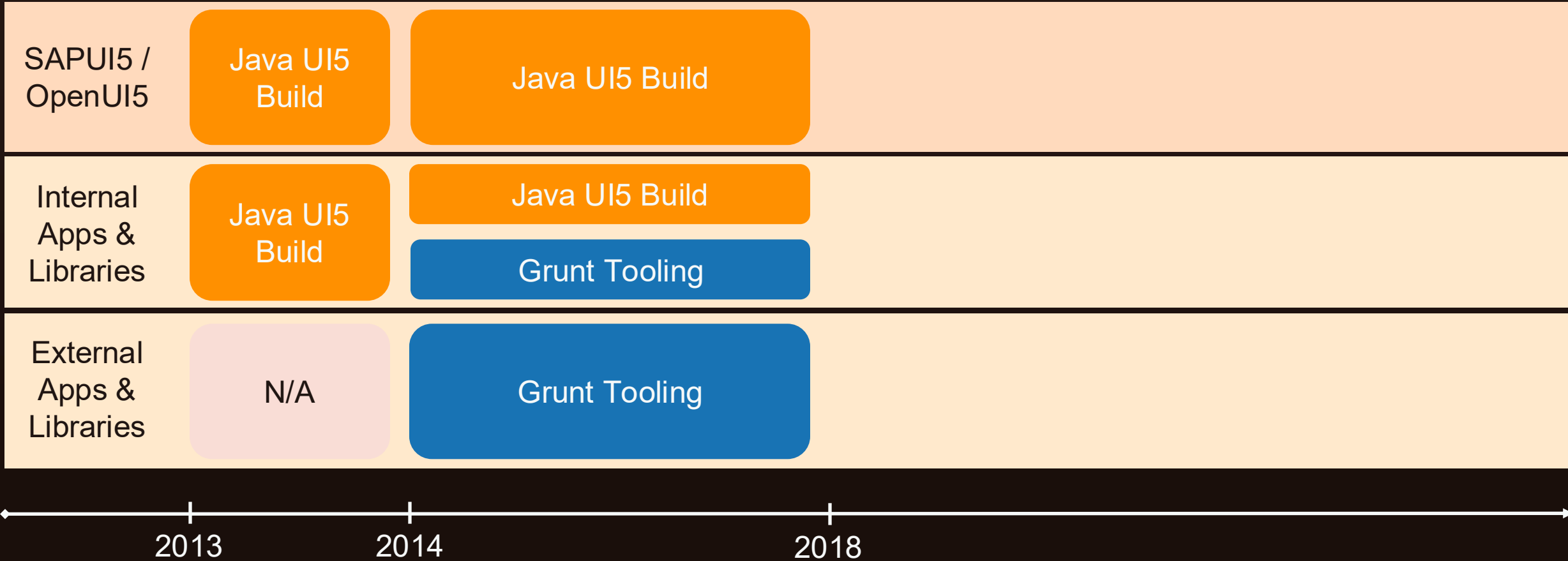
UI5 Tooling

Grunt Tooling

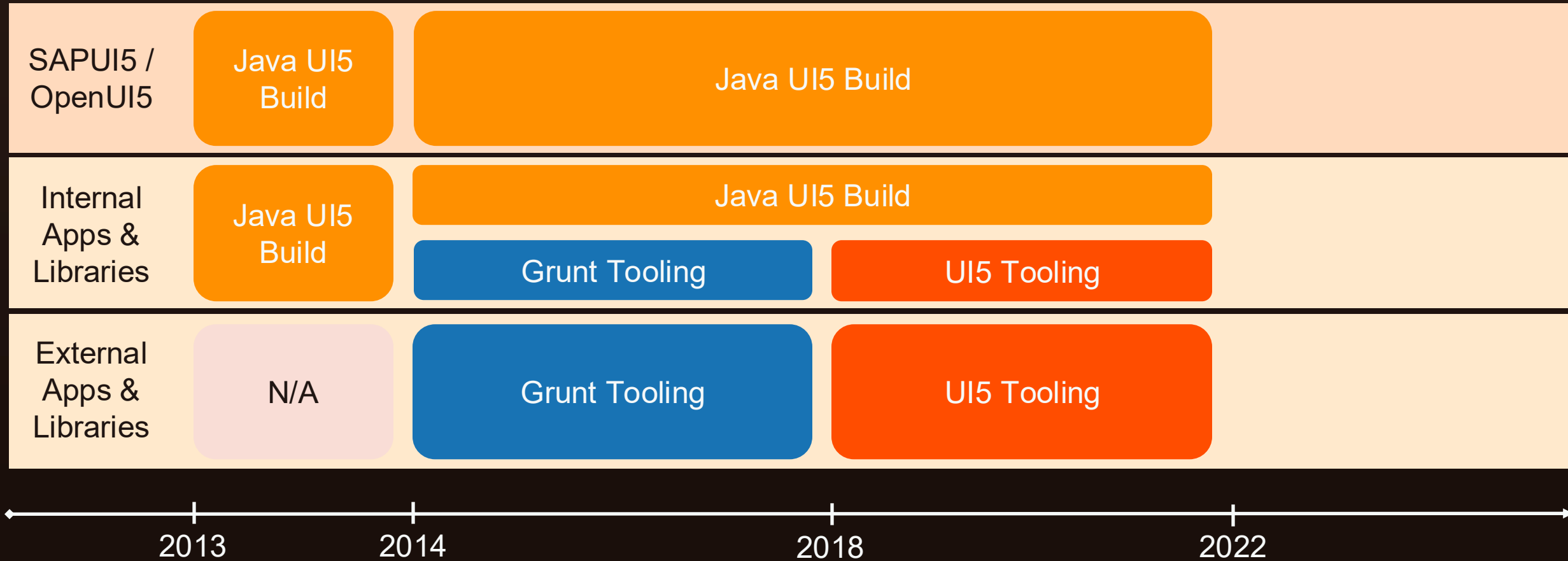
History of build tools for UI5



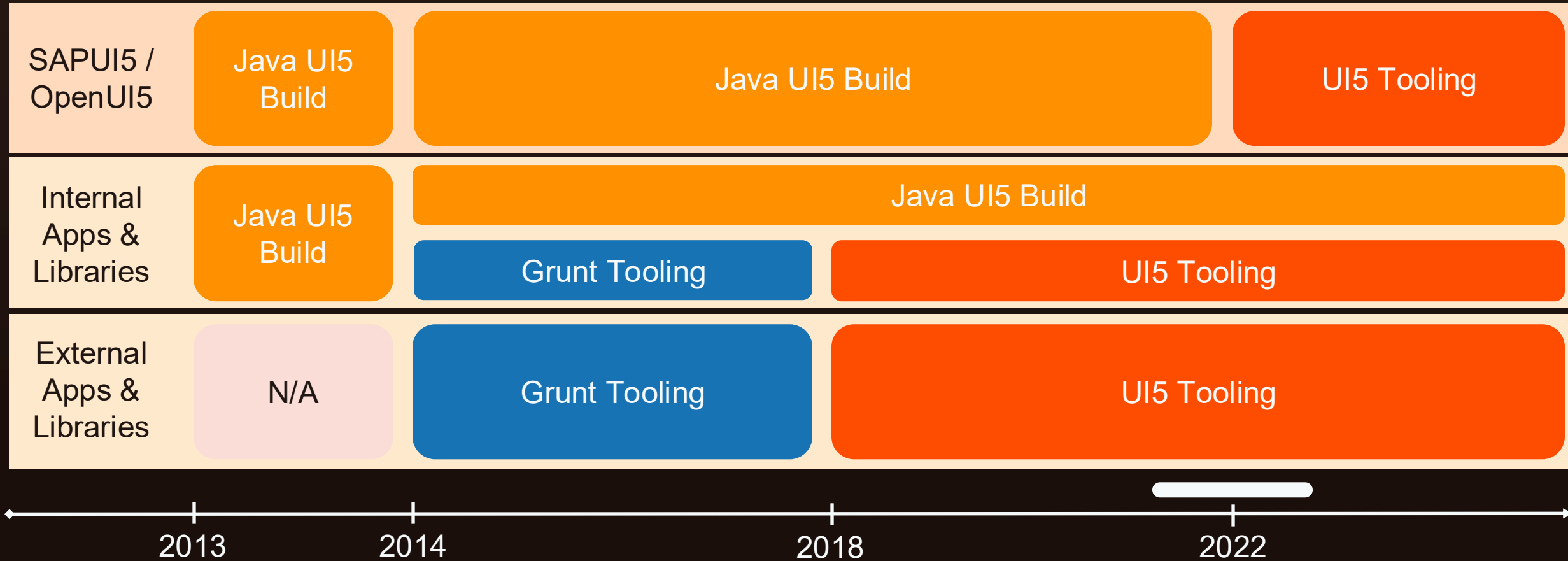
History of build tools for UI5



History of build tools for UI5



History of build tools for UI5



Build tools for UI5

Java UI5 Build

- Based on Java / Maven
- Full-fledged feature-set (w/ legacy)
- Proprietary / SAP internal



Grunt Tooling

- Based on the Grunt Task Runner
- Limited feature-set
- Open Source (Apache 2.0)



UI5 Tooling

- Based on Node.js (JavaScript)
- Full-fledged feature-set (w/o legacy)
- Open Source (Apache 2.0)



Build tools for UI5

Deprecated

Java UI5 Build

- Based on Java / Maven
- Full-fledged feature-set (w/ legacy)
- Proprietary / SAP internal



Grunt Tooling

- Based on the Grunt Task Runner
- Limited feature-set
- Open Source (Apache 2.0)



UI5 Tooling

- Based on Node.js (JavaScript)
- Full-fledged feature-set (w/o legacy)
- Open Source (Apache 2.0)



Reasons to migrate to UI5 Tooling



Double maintenance



Expertise in different technologies required



Old parsers limited to ES5 Syntax



Performance

Reasons to migrate to UI5 Tooling



Double maintenance



Expertise in different technologies required



Drinking our own champagne



Old parsers limited to ES5 Syntax



Performance

Migrating to UI5 Tooling

Java UI5 Build

- Based on Java / Maven
- Full-fledged feature-set (w/ legacy)
- Proprietary / SAP internal



UI5 Tooling

- Based on Node.js (JavaScript)
- Full-fledged feature-set (w/o legacy)
- Open Source (Apache 2.0)



The Challenge

Java UI5 Build

- Based on Java / Maven
- **Full-fledged feature-set** (w/ legacy)
- Proprietary / SAP internal



UI5 Tooling

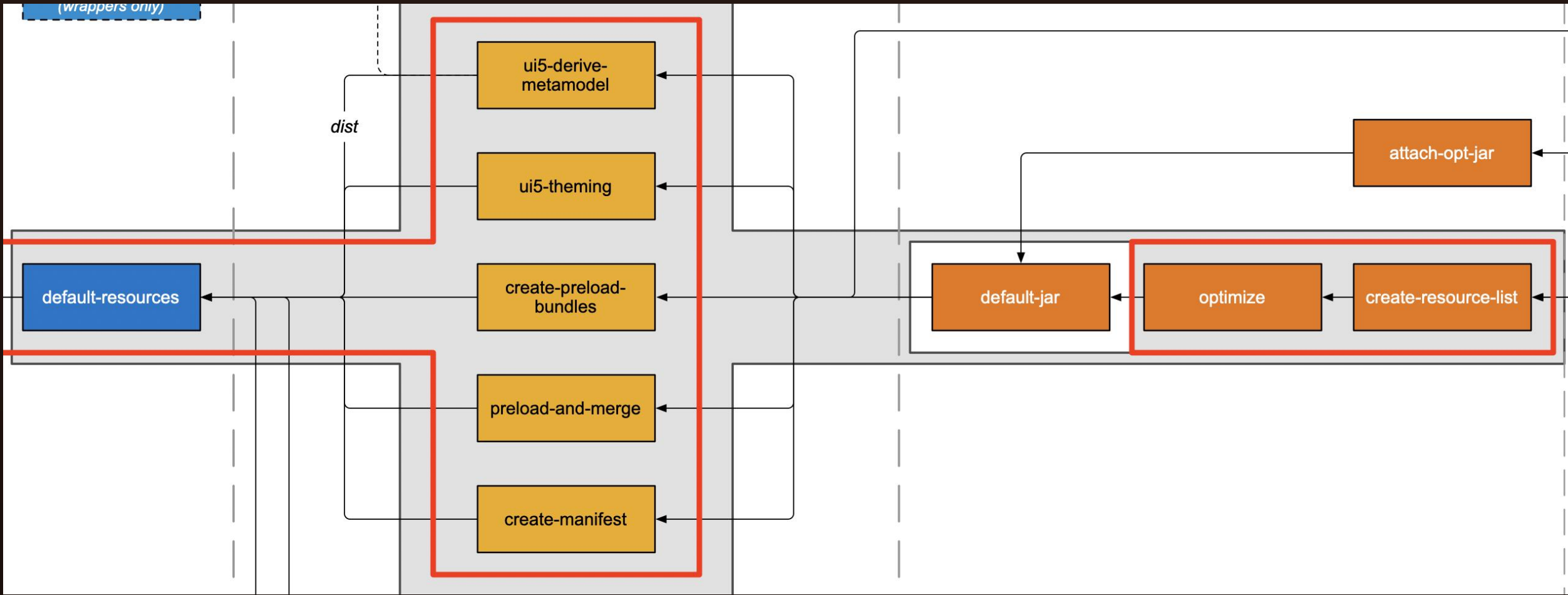
- Based on Node.js (JavaScript)
- **Full-fledged feature-set** (w/o legacy)
- Open Source (Apache 2.0)



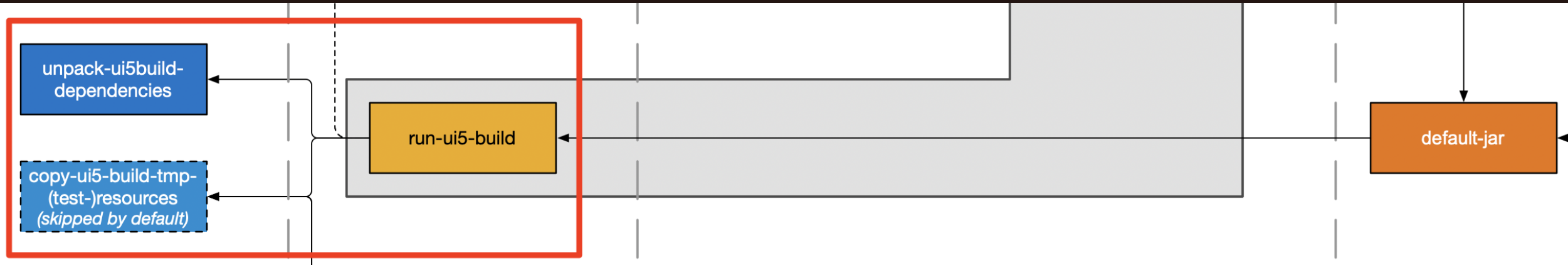
Is the build result
the same though?



Legacy build using "Java UI5 Build" tools

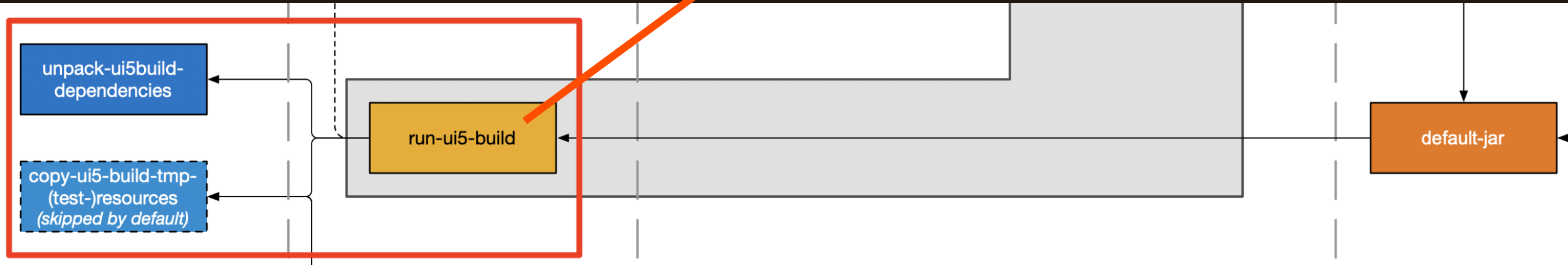


New build using UI5 Tooling



New build using UI5 Tooling

> ui5 build [...]



The road to feature parity

Java UI5 Build

UI5 Tooling
Build

The road to feature parity

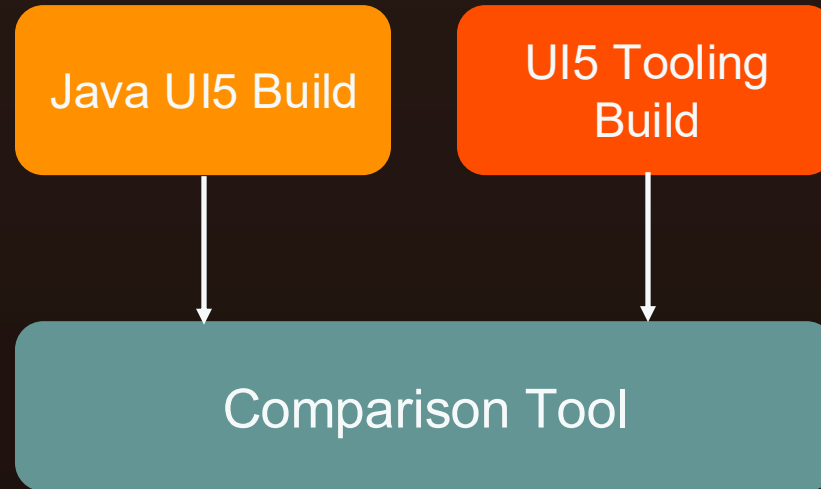
Java UI5 Build

UI5 Tooling
Build

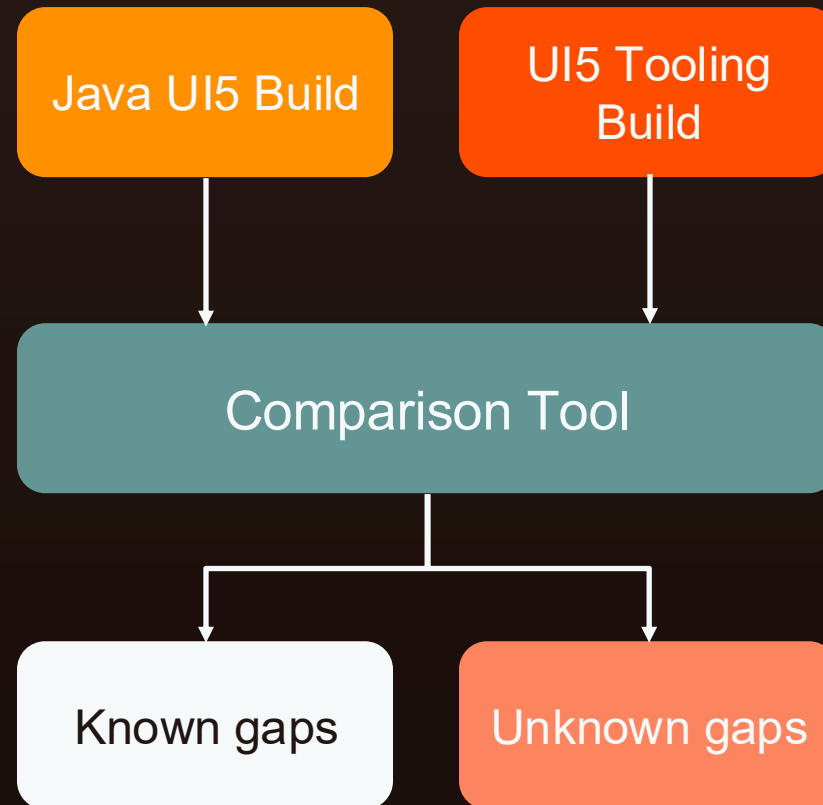
Is the build result
the same?



The road to feature parity



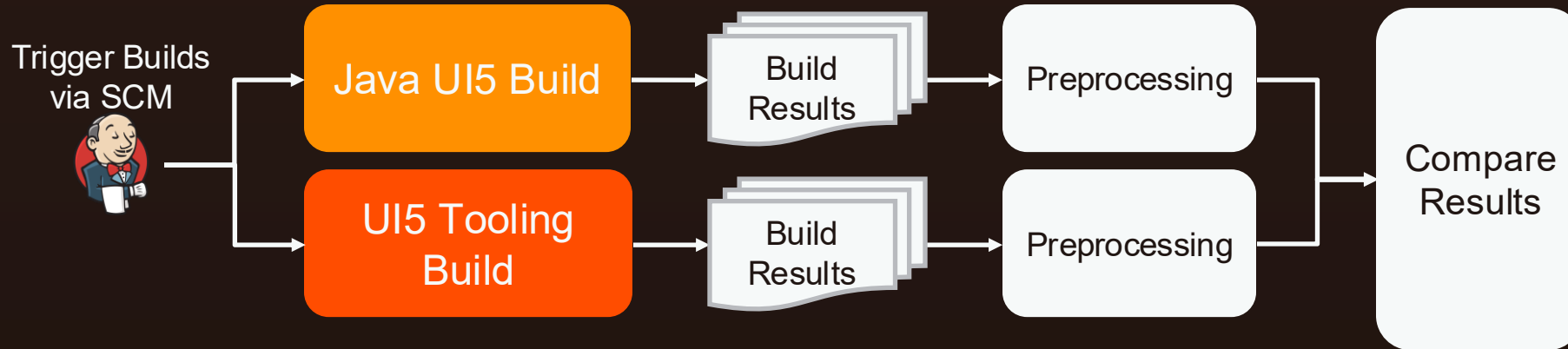
The road to feature parity



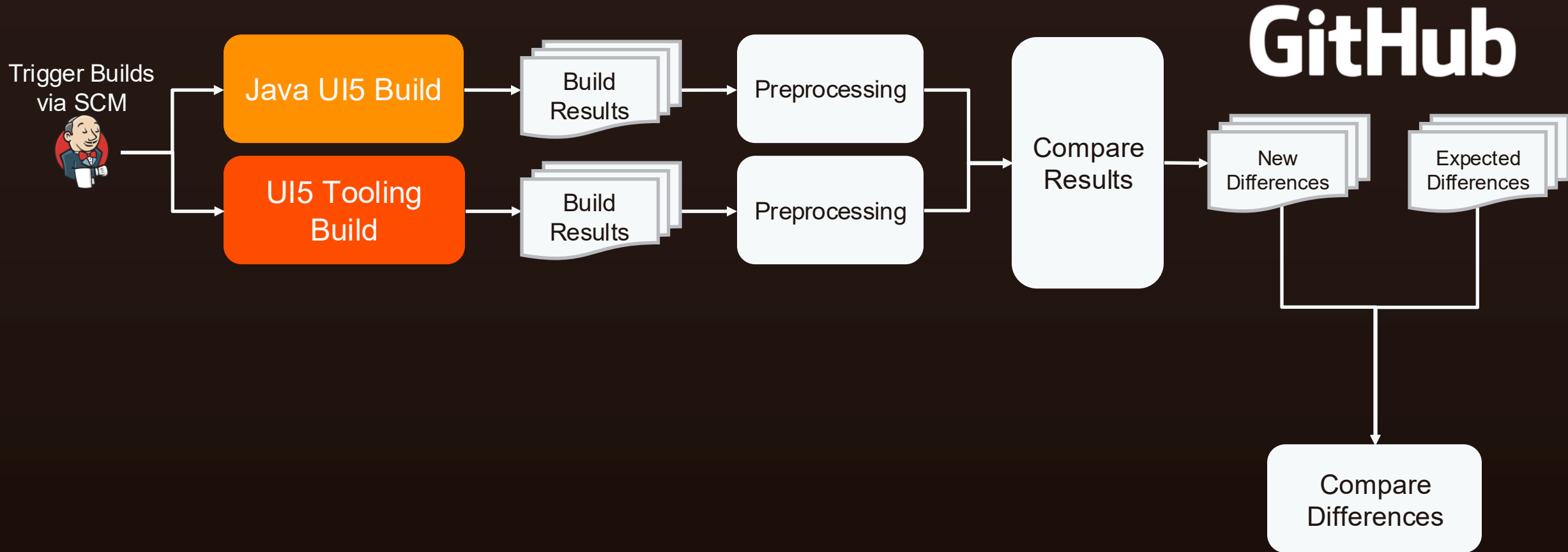
The Comparison Tool



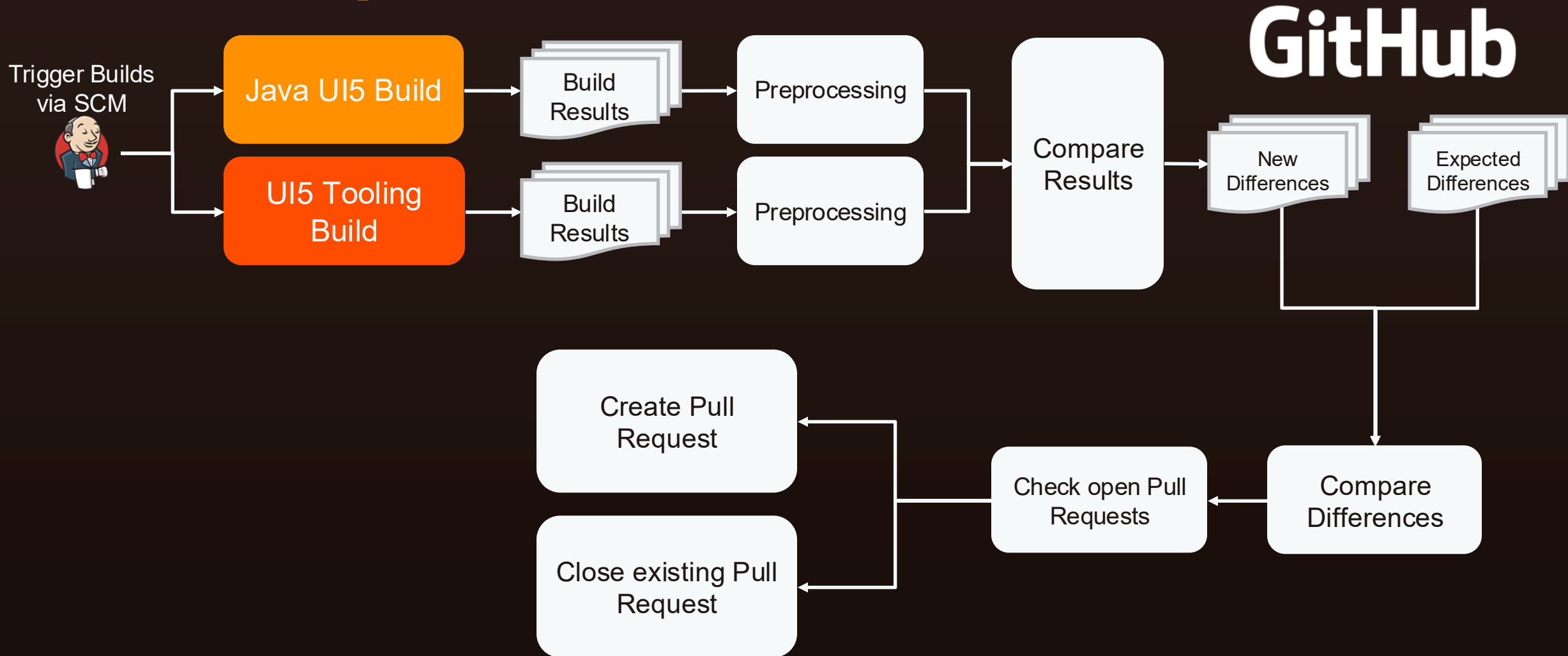
The Comparison Tool



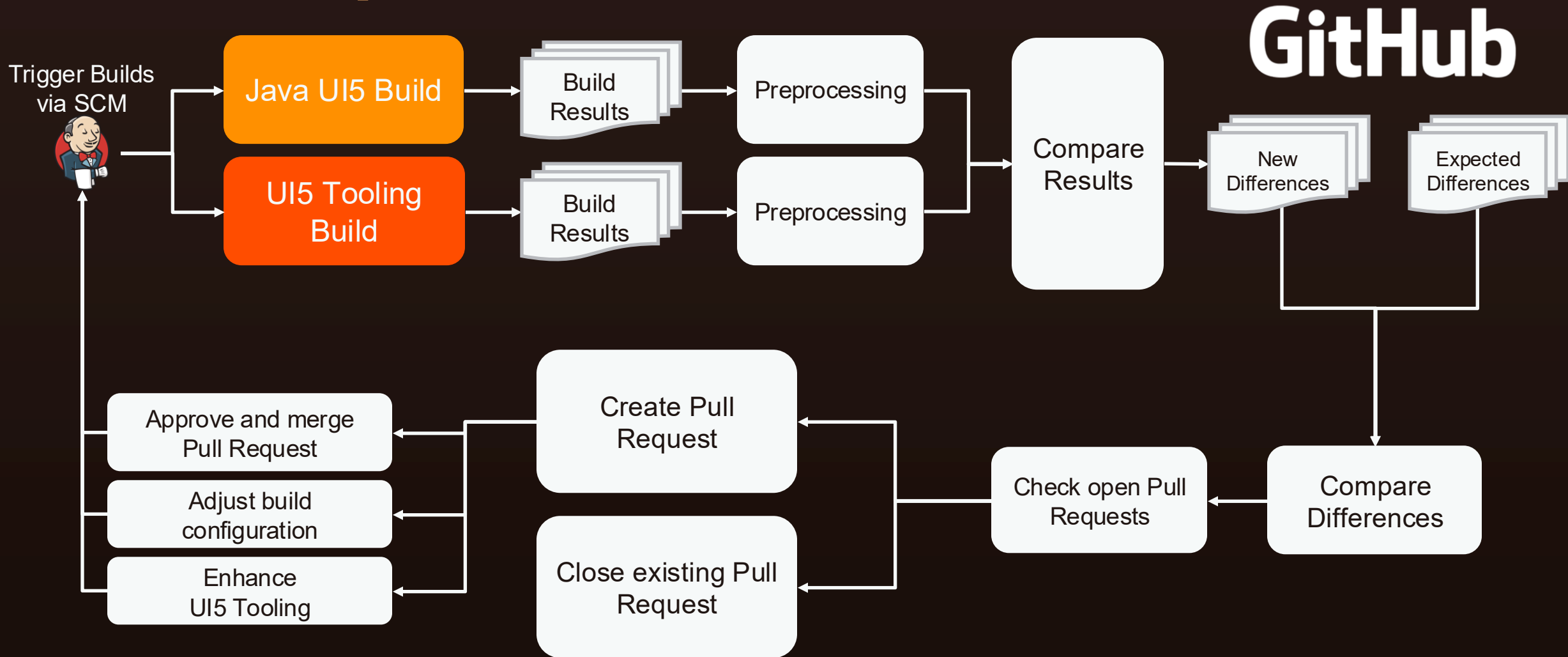
The Comparison Tool



The Comparison Tool



The Comparison Tool



Automated Pull Requests

openui5[5650430]: [INTERNAL] remove the Flexil window #762

 Merged Florian Vogt merged 1 commit into `master` from `update-ef2e0f65-fb6f-4a66-af20-49972a`

 Conversation **3**  Commits **1**  Checks **0**  Files changed **1**



ui5-core-founda... (UI5 Core Foundation Bot) commented on 15 Jul 2022 · edited ▾

Signature: 5b5afd417673f97f9ef8c7d269490dea7c6b556e

First detected with

[INTERNAL] remove the Flexibility section in the UI5 support window

Gerrit change: [openui5/+/5650430](#) (link to PS3)

POMahawk build: #49085

Latest POMahawk executions

- openui5 change 5663844: ⚠️ PS3 shows differences in build #49194
- sapui5/sapui5.runtime change 5664836: ⚠️ PS11 shows differences in build #49215

  Update snapshots based on openui5 change 5650430 ...



ui5-core-founda... (UI5 Core Foundation Bot) commented on 18 Jul 2022

The same difference appeared again for a different change:

[INTERNAL] sap.ui.model.odata.v4.ODataModel#getKeepAliveContext: obse

Gerrit change: [openui5/+/5663844](#) (link to PS3)

POMahawk build: #49194



ui5-core-founda... (UI5 Core Foundation Bot) commented on 18 Jul 2022

The same difference appeared again for a different change:

[FIX] comp.SmartVM: argument not passed to the inner VM control

Gerrit change: [sapui5.runtime/+/5664836](#) (link to PS11)

POMahawk build: #49215



✓ Florian Vogt approved these changes on 18 Jul 2022

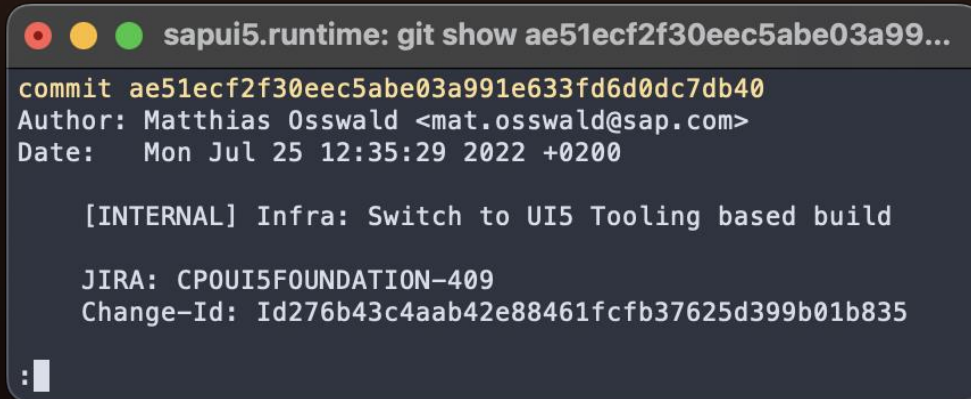
Florian Vogt (Florian Vogt) left a comment

LGTM



 Florian Vogt merged commit `6ff560e` into `master` on 18 Jul 2022

Making the move

A terminal window with a dark background and light gray text. The title bar shows three colored circles (red, yellow, green) followed by the text 'sapui5.runtime: git show ae51ecf2f30eec5abe03a99...'. The main content displays the output of a 'git show' command for a specific commit. The text is as follows:

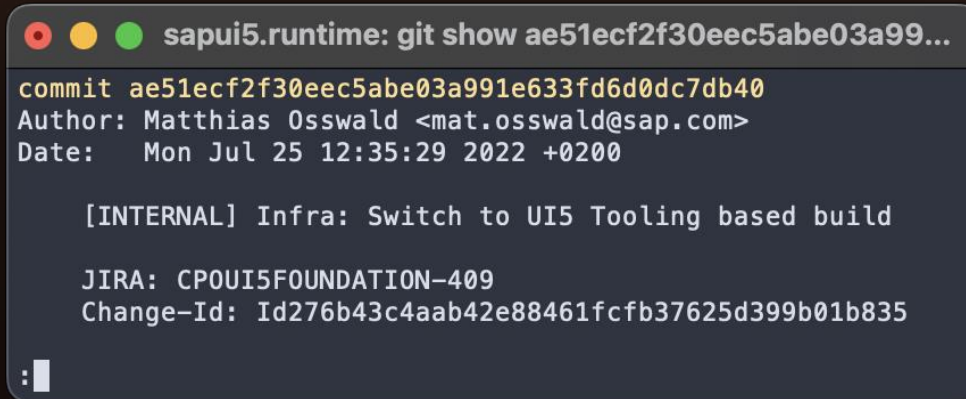
```
commit ae51ecf2f30eec5abe03a991e633fd6d0dc7db40
Author: Matthias Osswald <mat.osswald@sap.com>
Date:   Mon Jul 25 12:35:29 2022 +0200

    [INTERNAL] Infra: Switch to UI5 Tooling based build

    JIRA: CPOUI5FOUNDATION-409
    Change-Id: Id276b43c4aab42e88461fcfb37625d399b01b835
```

At the bottom left of the terminal window, there is a small icon consisting of a colon followed by a vertical bar (:|).

Making the move

A terminal window with a dark background and light gray text. The title bar shows three colored circles (red, yellow, green) followed by the text 'sapui5.runtime: git show ae51ecf2f30eec5abe03a99...'. The main content displays the output of a 'git show' command for a specific commit. The text is as follows:

```
commit ae51ecf2f30eec5abe03a991e633fd6d0dc7db40
Author: Matthias Osswald <mat.osswald@sap.com>
Date:   Mon Jul 25 12:35:29 2022 +0200

    [INTERNAL] Infra: Switch to UI5 Tooling based build

    JIRA: CPOUI5FOUNDATION-409
    Change-Id: Id276b43c4aab42e88461fcfb37625d399b01b835
```

A cursor is visible at the bottom left of the terminal window.

And it just worked.

Making the move

```
sapui5.runtime: git show ae51ecf2f30eec5abe03a99...  
commit ae51ecf2f30eec5abe03a991e633fd6d0dc7db40  
Author: Matthias Osswald <mat.osswald@sap.com>  
Date: Mon Jul 25 12:35:29 2022 +0200  
  
[INTERNAL] Infra: Switch to UI5 Tooling based build  
  
JIRA: CPOUI5FOUNDATION-409  
Change-Id: Id276b43c4aab42e88461fcfb37625d399b01b835  
:
```

And it just worked. 🥳

Lessons Learned



Comparison was game changer



"Fail" early and often



No severe issues after migration



Scope was hard to plan

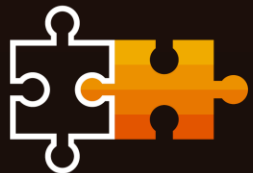
Further steps



UI5 Tooling integration in
SAP-internal application builds



Keep UI5 Tooling as slim as possible



Standard UI5 Tooling Extensions

github.com/SAP/ui5-tooling-extensions

UI5con

**Come meet us in
the Expert Corner!**

1:45pm to 3pm

Thank you!

Florian Vogt

SAP SE

E-mail: florian.vogt@sap.com

Github: flovogt

Matthias Oßwald

SAP SE

E-mail: mat.osswald@sap.com

Github: matz3

Merlin Beutlberger

SAP SE

E-mail: m.beutlberger@sap.com

Github: RandomByte